Homecoming

A discharge-to-home copilot for family caregivers

Nothing enters the care plan until a person confirms it.

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SPONSOR PROBLEM
ACL Caregiver AI Prize Challenge
Administration for Community Living, U.S. HHS

THE MOMENT

Discharge day is a handoff with no receiver

15–30

pages handed to a family member at the hospital door

Written for clinicians. Read by a daughter in a parking garage.

1. **New medications**

   Names, doses, and timing that did not exist last week.

2. **Changed and stopped drugs**

   The most dangerous line in the packet is the one that says stop.

3. **Follow-up appointments**

   Scattered across pages, often without dates attached.

4. **Warning signs**

   The symptoms that mean call now - buried in prose.

POSITIONING

A chatbot solves the wrong half of this

A general caregiver chatbot

- Answers questions faster
- Generates fluent, confident text
- Explains what a term means
- Improves efficiency
- Cannot tell you whether it made the answer up

It is also ACL's own listed use case so it is where most submissions will land.

Homecoming

- Takes one transition moment
- Treats it as a human-factors safety problem
- Every item traced to a page and a quote
- Prevents error, not just effort
- **Refuses to assert what it cannot ground**

Narrow scope, verifiable output.

HOW IT WORKS

Packet in, verified plan out

1
Upload
PDF, pasted text, or a photo

→

2
Extract
Claude returns structured items

→

3
Ground
Verbatim quote must be found

→

4
Confirm
A human promotes each item

→

5
Live plan
Reminders, fridge sheet, family link

Steps 3 and 4 are the product.

Everything else is plumbing that other teams will also have. The two gates are what make the output trustworthy enough to act on.

Quote grounding - hallucination blocked in code

Every item Claude returns must carry:

- a verbatim quote from the source document
- the page number it came from
- a structured payload matching a fixed schema

The function then searches the submitted page text for that exact quote.

`extract.mjs`

If the quote is not found

the item is discarded

before it ever reaches the database.

Not a warning. Not a low score. Deleted.

Anti-hallucination as a code path, not a prompt instruction.

Asking a model politely not to invent things is a hope. Verifying its output against the source is a guarantee.

Human confirmation - enforced by the schema

`candidate_items`

Unconfirmed AI output.
Visible to the caregiver for review.
Never surfaced as care instructions.

a person
confirms

→

`plan_items`

The confirmed care plan.
Drives reminders, the fridge sheet,
and the family share link.

`audit_log`

Append-only.
Every confirm,
edit, reject,
remove.

Two tables, not a boolean flag

A confirmed column can be defaulted, mis-set, or bypassed by one careless query. Promotion across tables has to be written deliberately.

This is also the prompt-injection defense

A malicious instruction hidden in a document must survive quote verification and a human reading it and clicking confirm.

THE HONEST EDGE CASE

When there is nothing to ground against

Caregivers photograph things. A nurse's handwritten note on the back of a page is often the most important instruction in the packet - and it has no machine-readable source text to verify a quote against.

Photos are still supported

Refusing image upload would fail the real user. Claude reads the image and extracts what it can.

But Gate 1 cannot run

There is no submitted page text to search. The grounding check has nothing to check against.

So they are force-flagged

Image-derived items are marked NEEDS REVIEW and can never be bulk-confirmed. Every one is read individually.

Validated against a handwritten note: the system extracted the content while correctly refusing to claim confidence in it.

LIVE DEMO

Watch for four things

1. **The citation on every card** — Page number and source quote, on each extracted item.
2. **The amber NEEDS REVIEW flag** — Low-confidence and image-derived items, visually separated.
3. **The confirm click** — The moment an item crosses from AI output into the care plan.
4. **The fridge sheet** — One printed page that ends up taped to a refrigerator door.

ARCHITECTURE

Every secret and every check lives server-side

CLIENT

- React 18 + Vite
- Tailwind CSS
- pdfjs-dist page extraction
- Anon key only

→

SERVERLESS · Netlify Functions

- extract.mjs — grounding gate
- guidance.mjs — RAG retrieval
- ask.mjs — plan-scoped Q&A
- share.mjs — revocable link
- send-reminders.mjs — scheduled

→

DATA · Supabase

- Postgres + RLS on every table
- Auth
- Private storage bucket
- pgvector for guidance

The Anthropic key never reaches the browser

Only VITE_-prefixed variables ship to the client. Model calls, service-role access, and email all run inside functions.

Authorization lives in the database

Row Level Security keyed on care-circle membership not client-side route guards that a devtools console can walk around.

STATUS

Built, deployed, and running today

✓ Grounded extraction

PDF, pasted text, and photo ingestion with the quote-verification gate.

✓ Review and confirm

Confirm, edit, reject, remove — with confidence flags and an append-only audit log.

✓ Printable fridge sheet

The confirmed plan as one page, formatted for a refrigerator door.

✓ Family share link

Revocable, read-only view for relatives who are not managing care.

✓ RAG guidance

Answers only from a curated corpus — otherwise it declines and says ask your care team.

✓ FHIR export

The confirmed plan as a standards-compliant bundle, ready to hand to a clinician.

Deployed at homecoming-care.netlify.app · auth, Row Level Security, and rate limiting live in production

SAFETY POSTURE

What this system refuses to do

No diagnosis, dosing, or triage

Claude may only restructure what the document already says. Enforced in the system prompt and validated on output.

No ungrounded answers

If no trusted source clears the relevance threshold, guidance declines and tells the caregiver to ask their care team, rather than falling back on model knowledge.

No real patient data, ever

All development, testing, screenshots, and demo content use one synthetic discharge summary. No real person's medical documents were used at any point.

Homecoming organizes discharge instructions. It is not medical advice and it is not a medical device.

WHAT COMES NEXT

From verified plan to validated product

NEXT

Caregiver interviews

5–8 structured interviews to trace design decisions back to lived experience — the strongest remaining gap.

IN PROGRESS

Email reminder delivery

Schema and scheduled function are built; production delivery and deliverability testing remain.

PLANNED

Prompt-injection test suite

Adversarial documents run as automated tests, so the injection defense is proven rather than argued.

PLANNED

Clinical review

Bring a discharge nurse in to challenge the four item categories and the warning-sign extraction.

Every item in a caregiver's plan can be traced to the page it came from and a person put it there.

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